

## **Tutorial 2 – Advanced String Manipulation**

1. Given a phone number 60123456789 (as int64\_t type for integer of 64 bits)
  - (a) Declare the phone number as integer and convert it to string value
  - (b) Display the string as +60 12-3456789 (Hint: Use concatenate and sub-string)
2. Given the string "45.67, 23, 12.5, 10, 78.9". Write a C++ program to find and display the sum of these five values.

*Note: You need to use stringstream, a loop, getline and stod functions to read and convert the string value to floating point values. The getline function can be used with delimiter (',') to split the string.*
3. Given the string "Two dimensional array", find and display the index of first and last vowel located in the string
4. In a greeting message like this "henry ford, good morning! Welcome to the HILTON HOTEL.". You are asked to use insert() and transform() functions to transform and display the message as this "Mr. HENRY FORD, good morning! Welcome to the Hilton Hotel.".
5. In a monthly sales reporting, you are to enter the name of the sales team and the month of reporting as follows. The sales made by the sales team has been compiled and tabulated as follows:-

|              |             |
|--------------|-------------|
| Team: Great  | Month: June |
| Member       | Sales (RM)  |
| -----        | -----       |
| Howard Goh   | 68950.35    |
| Carrie Hank  | 46900.20    |
| Darren Smith | 72195.75    |

You are required to declare and initialize these values in 2 vector arrays. Use string formatting functions to generate the output above taking the values from user input and the 2 arrays.

6. A postpaid telco customer received a text message "Bill amount: RM128.50". To confirm the customer of the exact payment to make, you are asked to write a C++ program segment to generate this message:

"Please make the payment RM128.50 by the due date given in the bill".

*Note: You are to derive a suitable pattern and use search regex function*

7. Prompt the user to enter an IP address. Write a C++ program segment to validate it. Example of a valid IP address is like 192.168.10.2. Each of the 4 numbers can have 1 to 3 digits. Display result of the validation.

*Note: You are required to derive a suitable pattern and use match regex function*